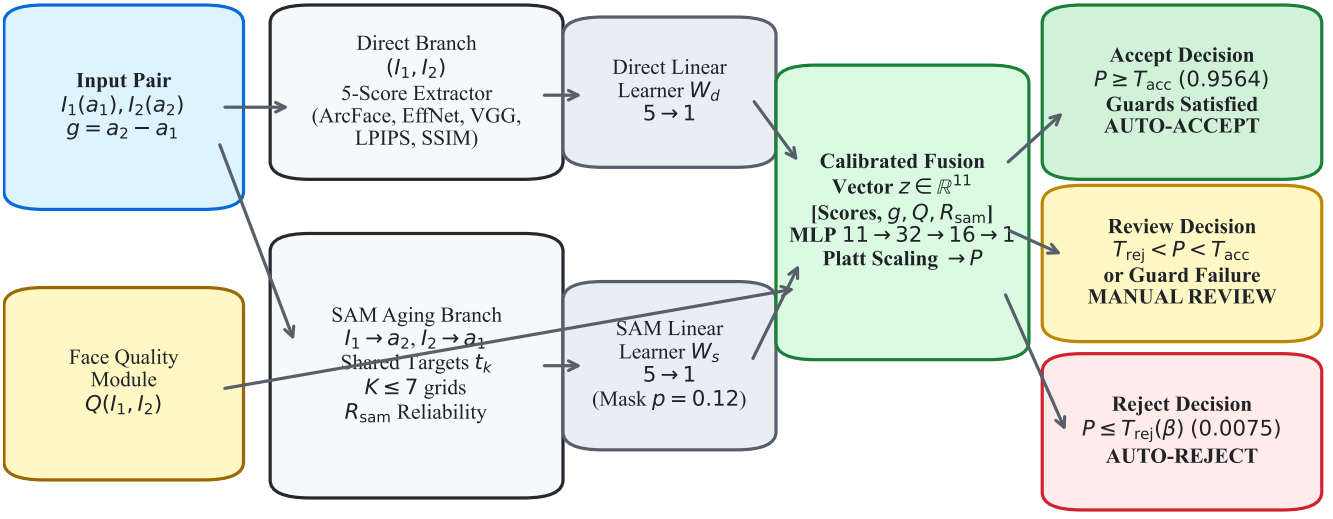

Input Pair
$I_1(a_1), I_2(a_2)$
$g = a_2 - a_1$
Face Quality Module
$Q(I_1, I_2)$
Direct Branch
(I_1, I_2)
5-Score Extractor
(ArcFace, EffNet, VGG, LPIPS, SSIM)
SAM Aging Branch
$I_1 \to a_2, I_2 \to a_1$
Shared Targets t_k
$K \le 7$ grids
R_{sam} Reliability
Direct Linear Learner W_d
$5 \to 1$
SAM Linear Learner W_s
$5 \to 1$
(Mask $p = 0.12$)
Calibrated Fusion
Vector $z \in \mathbb{R}^{11}$
[Scores, g, Q, R_{sam}]
MLP $11 \to 32 \to 16 \to 1$
Platt Scaling $\to P$
Accept Decision
$P \ge T_{\text{acc}}$ (0.9564)
Guards Satisfied
AUTO-ACCEPT
Review Decision
$T_{\text{rej}} < P < T_{\text{acc}}$
or Guard Failure
MANUAL REVIEW
Reject Decision
$P \le T_{\text{rej}}(\beta)$ (0.0075)
AUTO-REJECT